



XGBoost

supervised

Isolation Forest

anomaly

CRF

sequence (PGM)

SHAP

explainability

CSLAB CHALLENGE 2026

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CIC-IDS-2017 · WEB ATTACK DETECTION

Detecting Web Attacks with a Three-Layer Machine Learning System

Brute Force · XSS · SQL Injection generated against DVWA

THE PROBLEM & THE DATA

Rare attacks hidden in normal traffic

Dataset: CIC-IDS-2017 (GeneratedLabelledFlows — keeps timestamps & IPs).

170,231 clean flows × **77 features** (duration, packets, bytes, timing, TCP flags & window sizes).

Only 1.28% are attacks — severe imbalance. Guessing “benign” always scores 98.7% accuracy yet catches zero attacks.

→ Evaluate with **F1 & Recall**; balance training with SMOTE. Metadata (IPs, ports) dropped so the model learns behaviour, not attacker identity.

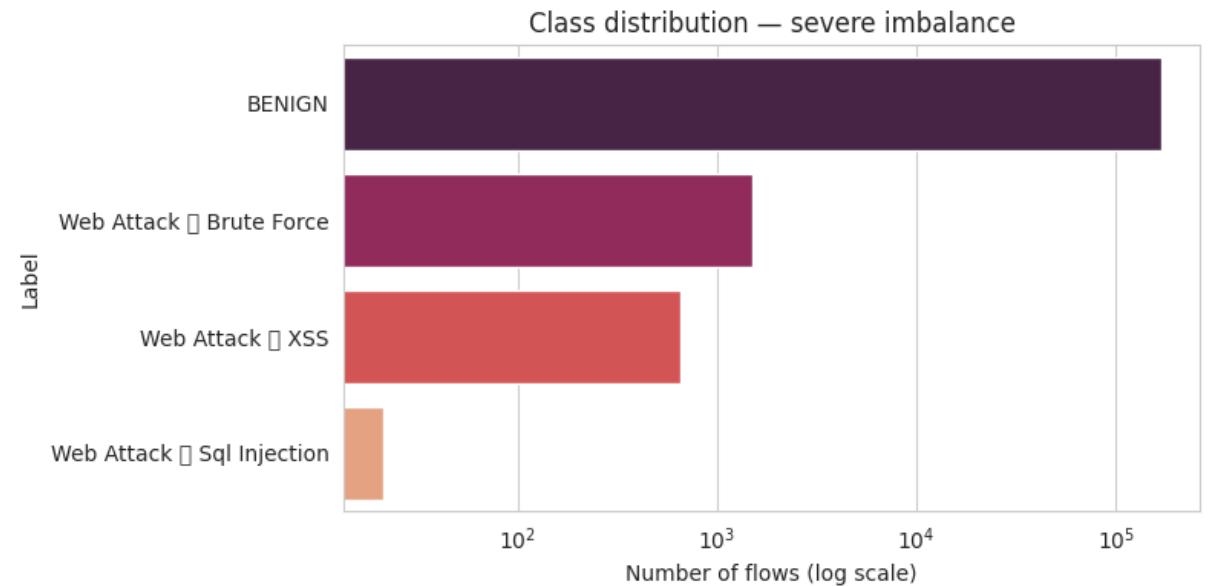


Figure 1 — class distribution (log scale): 168,051 benign vs 2,180 attacks.

Three complementary detection layers

01

XGBoost

Supervised · per-flow

300 boosted trees, each fixing the previous ones' errors. Learns known web-attack patterns from labels.

02

Isolation Forest

Unsupervised · per-flow

Flags flows that isolate quickly under random cuts — a safety net for novel anomalies, no labels needed.

03

CRF

Sequence · PGM

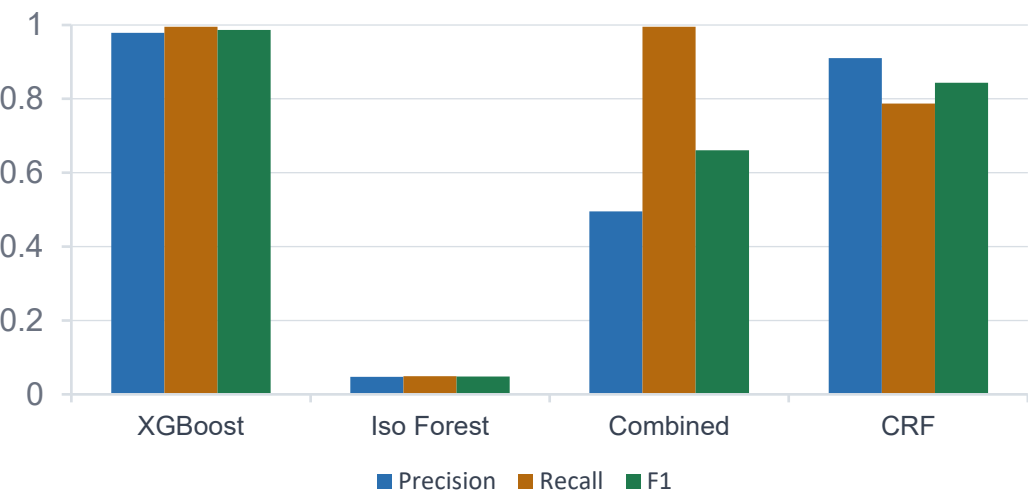
A probabilistic graphical model that labels each flow using its time-ordered neighbours — detects attack sequences.

Explainability: SHAP reveals which network features drive each detection.

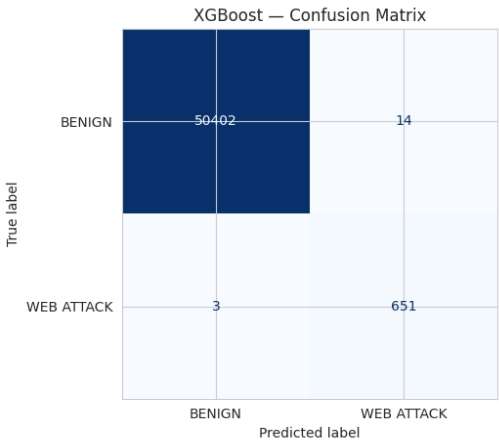
Pipeline: clean → scale → SMOTE (train only) → stratified split for per-flow models, blocked split for the CRF — a leak-free evaluation.

RESULTS

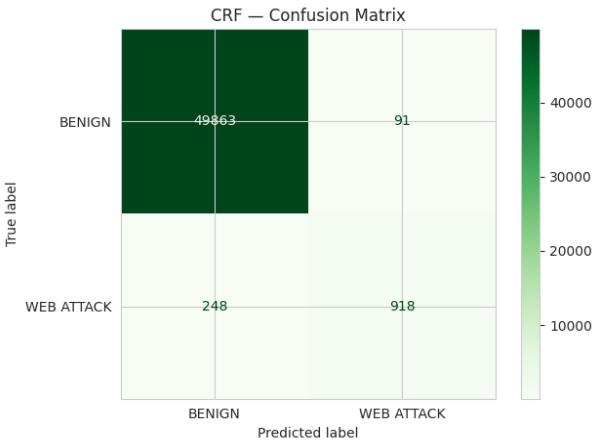
Detection performance



Model comparison — Precision / Recall / F1



XGBoost confusion matrix



CRF confusion matrix

0.987

XGBoost F1

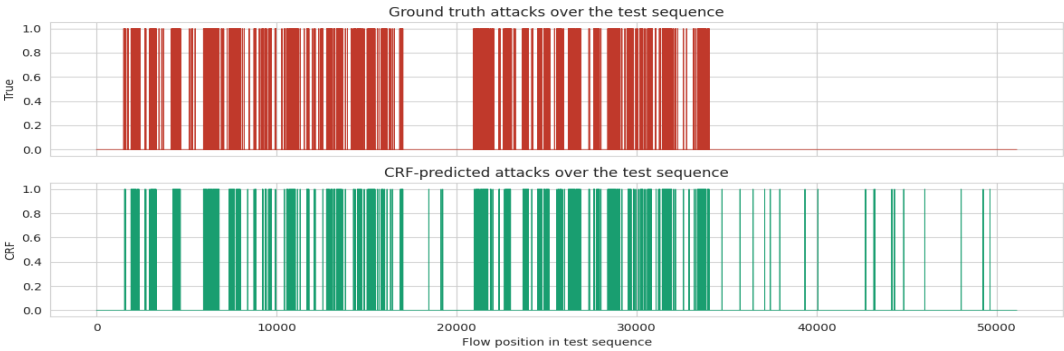
1.000

ROC-AUC

0.844

CRF F1

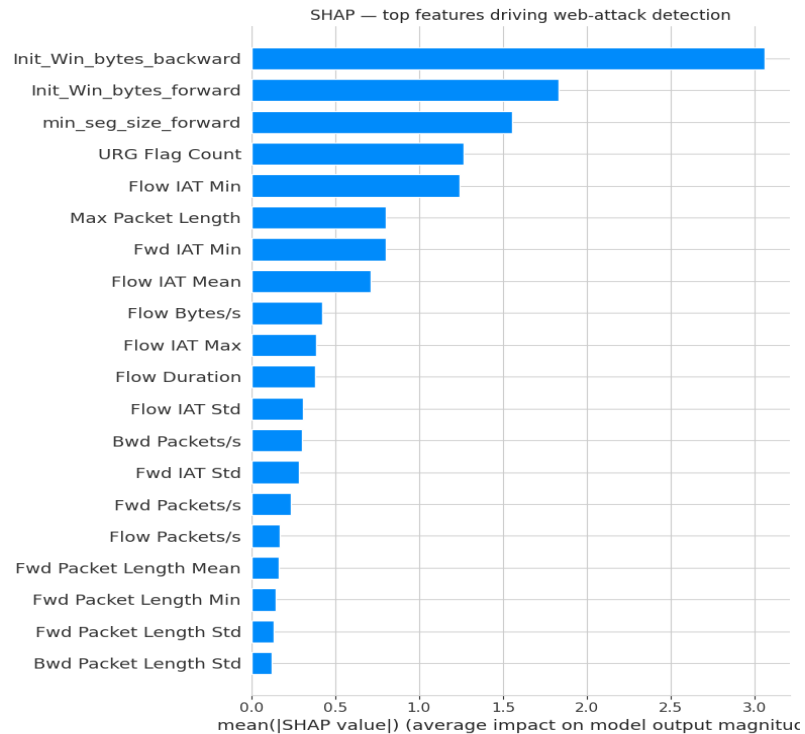
XGBoost caught 651/654 attacks (missed 3) with 14 false alarms. Isolation Forest is weak alone (expected). CRF detects attacks in temporal sequence.



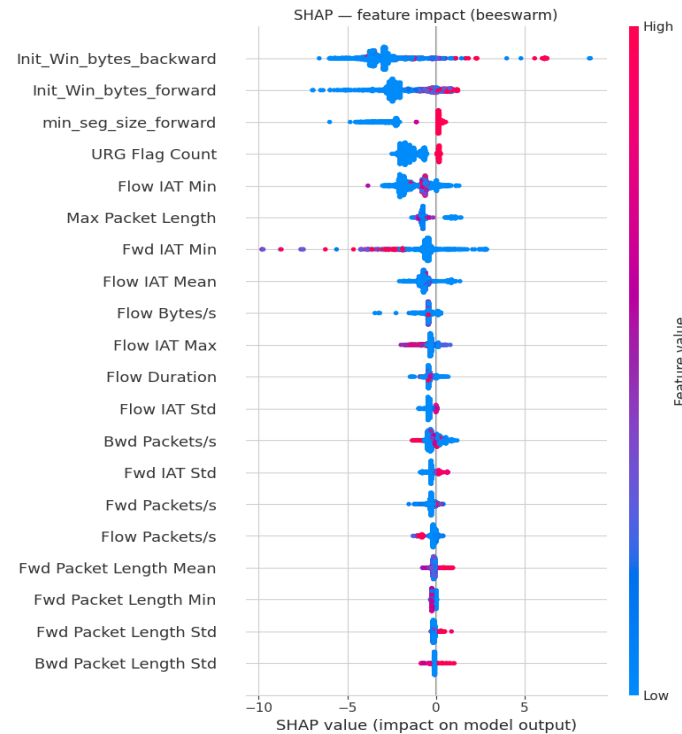
CRF detections vs ground truth over the time sequence

EXPLAINABILITY & CONCLUSIONS

Why it works — and what we found



SHAP — top features by importance



SHAP beeswarm — right = pushes toward attack

Key findings

- ✓ XGBoost detects known attacks near-perfectly (F1 0.987)
- ✓ Isolation Forest adds novelty detection (no labels)
- ✓ CRF models attack sequences in time (F1 0.844)
- ✓ CRF uses neighbouring flows to detect attack bursts
- ✓ Attacks signalled by TCP window sizes, timing & flags
- ✓ Deployed as a live Streamlit detection console